प्रदेश लोक सेवा आयोग, कर्णाली प्रदेश,
इन्जिनियरिङ्ग सेवा, सिभिल समूह, (जनरल, इरिगेशन, स्यानिटरी, हाईड्रोपावर) उप समूह, अधिकृतस्तर
सातौं तह, इन्जिनियर पदको खुला प्रतियोगितात्मक लिखित परीक्षा

चरण: प्रथम	परीक्षा मिति: २०८१/०४/१६	
समय: ३ घण्टा	पत्र: द्वितीय	पूर्णाङ्क: १००
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Section-A

1. **Build Back Better is the principle of rebuilding. What could be the principle of replanning of the post disaster settlements? [5]**

Answer:

Build Back Better (BBB) principle emphasizes rebuilding infrastructure, communities, and livelihoods after a disaster in a way that reduces future risks and improves resilience. Build Back Better (BBB) is the guiding principle for rebuilding after a disaster. Similarly, the principle of replanning post-disaster settlements focuses on creating safer, more sustainable, and disaster-resilient communities rather than simply restoring the previous layout. These principles emphasize that post-disaster recovery should not simply restore the previous settlement pattern but should improve it by reducing future risks and enhancing quality of life.

Principles of Replanning Post-Disaster Settlements:

i. Risk-informed land use planning:
   - Avoid rebuilding in high-risk areas such as floodplains, landslide-prone slopes, and fault zones.
   - Use hazard and vulnerability maps to guide future development.

ii. Build community resilience:
   - Design settlements to withstand future disasters.
   - Provide redundancy in critical infrastructure and evacuation routes.

iii. Sustainability:
   - Protect natural ecosystems such as forests, wetlands, and river corridors.
   - Encourage environmentally friendly construction and resource-efficient infrastructure.

iv. Community participation:
   - Involve local communities in planning and decision-making.
   - Respect local culture, traditions, and livelihood needs.

v. Adaptive and flexible planning:
   - Allow for future expansion and changing hazard scenarios due to climate change and urban growth.
   - Periodically review and update settlement plans.

Section-B

2. **Determine the moment of resistance for rectangular section with:**
F_{ck}= 25 N/mm^2 ; F_y =500 N/mm^2 ; Tension reinforcement= 3 Nos 16 mm, width of section= 300 mm; Depth of section=350 mm; Depth of section=350 mm; Assume suitable cover. [5]

Answer:

Characteristic compressive strength of concrete: $f_{ck} = 25$ N/mm^2

Yield strength of steel: $f_y = 500\ \text{N/mm}^2$

Width of section: $b = 300\ \text{mm}$

Overall depth: $D = 350\ \text{mm}$

Tension reinforcement: $3\phi - 16$ mm

Assume clear cover = 25 mm

Step 1: Find the effective depth:

Diameter of bar= 16 mm

d= $D - \text{cover} - \frac{\phi}{2}d$ = $350 - 25 - 8 = \mathbf{317\ mm}$

Step 2: Area of steel

Area of one 16 mm bar

A= $\frac{\pi}{4}(16)^2 = 201.06\ \text{mm}^2$

For three bars, $A_{st} = 3 \times 201.06 = 603.2\ \text{mm}^2$

Step 3: Natural Axis Depth

For Fe500 steel,

$$0.36 f_{ck} b x_u = 0.87 f_y A_{st}$$

$$= 0.87 fyAst0.36(25)(300)xu$$

$$= 0.87(500)(603.2)0.36(25)(300)x_u$$

$$= 0.87(500)(603.2)0.36(25)(300)xu$$
$$= 0.87(500)(603.2)2700xu$$
$$= 2623922700x_u = 97.2\ \text{mm}xu$$
$$= 97.2\ mm$$

Step 4: Check limiting neutral axis

For Fe500,

$$x_{u,max} = 0.46d$$

$$= 0.46(317)$$

$$= 145.8\ mm$$

$97.2 < 145.8$ mm the section is **under-reinforced**.

Step 5: Moment of Resistance

$$M_u = 0.87 f_y A_{st}(d - 0.42 x_u)$$

$$= 0.87(500)(603.2)(317 - 0.42 \times 97.2)$$
$$= 0.87(500)(603.2)(317 - 0.42 \times 97.2)$$

Lever arm: $317 - 40.82 = 276.18\ mm$

Hence,

$$M_u = 262392 \times 276.18 Mu = 262392 \times 276.18 Mu = 72.48 \times 106\ N - mm M_u$$
$$= 72.48 \times 10^{6\ \text{N-mm}} Mu$$

$$\boxed{M_u = 72.5 \text{ kN-m (approximately)}}$$

3. **Give Courbon's assumption for cross girder in a bridge. Explain what is the conditions to be fulfilled, so that the assumption could be applied for the analysis. [5]**

Answer:

Courbon's method is used to determine the distribution of wheel loads among the longitudinal girders of a bridge deck.

The method is based on the following assumptions:

i. Cross girders (diaphragms) are infinitely rigid in the transverse direction.
ii. Longitudinal girders are identical and equally spaced.
iii. The bridge deck acts as a rigid floor system in the transverse direction.
iv. The longitudinal girders carry the load primarily by bending.
v. The load is transmitted to the girders through the rigid cross girders.
vi. The bridge is simply supported.
vii. Deformations are within the elastic limit.

The conditions that have to be fulfilled so that the assumption could be applied for the analysis are as follows:

i. Cross girders should be sufficiently rigid.
ii. Longitudinal girders should be parallel and equally spaced.
iii. Longitudinal girders should have equal flexural rigidity.
iv. Bridge deck should be symmetrical.
v. Bridge should be simply supported.
vi. Span-to-width ratio should be sufficiently large.
vii. Cross girders should be connected monolithically to the deck slab and longitudinal girders.
viii. Torsional effects should be negligible.

4. **List out the different methods of soil improvement techniques. Explain in detail stone column and sand pile soil improvement techniques and their applicability.[5]**

Answer:

Soil improvement techniques are those techniques that are used to enhance the engineering properties of soil, such as its strength, bearing capacity, stiffness, permeability, and durability**.** It is also known as soil stabilization techniques.

The different methods of soil improvement techniques are:

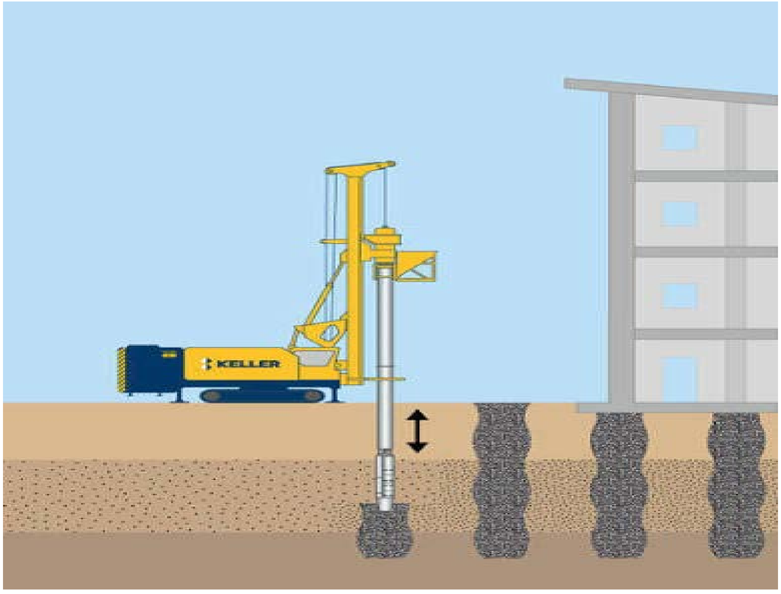
i. Mechanical improvement methods: compaction, blasting, vibro-compaction etc.
ii. Hydraulic improvement methods: preloading, sand drains, dewatering using well points
iii. Chemical stabilization methods: lime stabilization, cement stabilization, bitumen stabilization.
iv. Grouting Techniques: cement grouting, compaction grouting, jet grouting etc.
v. Reinforcement techniques: geotextiles, geogrids, geocells etc.
vi. Replacement methods: stone columns, sand compaction piles etc.
vii. Biological method: vegetative stabilization

Stone column:

A stone column is a vertical column of compacted crushed stone or gravel installed in soft ground to improve its engineering properties.

It is also known as vibro-replacement because weak soil is replaced by compacted stone. T improved the soil by:

i. Increasing the load-bearing capacity.
ii. Densifying the surrounding soil.
iii. Reducing total and differential settlements.
iv. Increasing resistance to liquefaction during earthquakes.



The diameter of the stone column ranges from 0.6-1.2 m and depth up to 20-25m depending on the equipment and soil condition. The spacing between the stone columns is kept 2-3 times the column diameter.

Applicability:

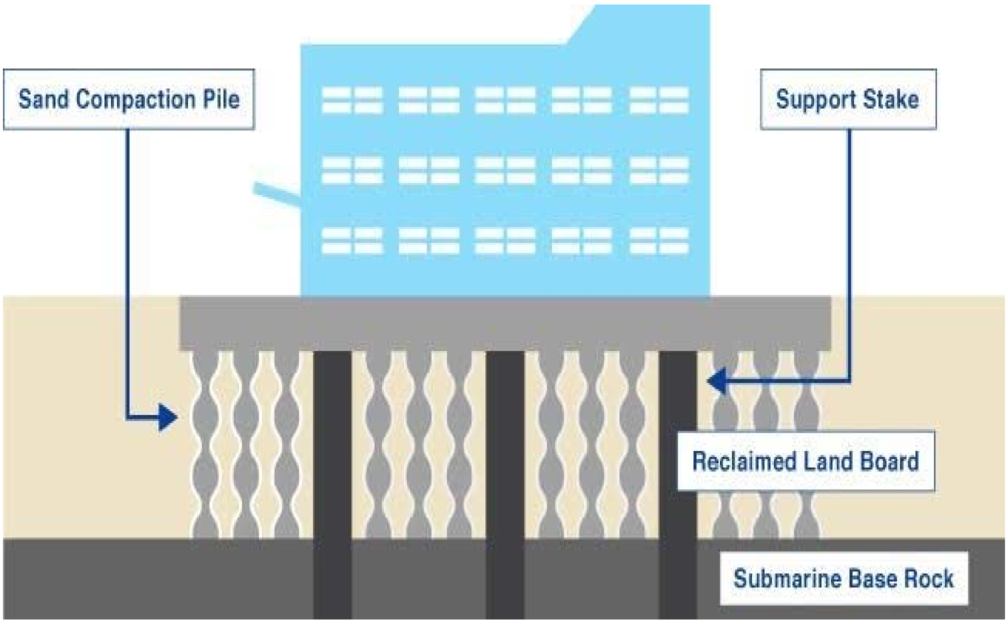
It is suitable for soft clay, silty clay, reclaimed land, embankments, highway and railway foundations, industrial buildings, airport runways and so on.

Sand pile:

It is a vertical column made of compacted sand installed in weak soil to improve strength and drainage. It is also called sand compaction pile.

It improves the soil by:

i. Densifying the loose soil.
ii. Replacing the part of weak soil with compacted sand.
iii. Providing drainage paths for pore water.
iv. Increasing shear strength and bearing capacity.



The diameter of sand pile ranges from 0.3-0.8 m and depth from 10-20 m. The spacing of the sand column is kept as 2- times the pile diameter.

Applicability: It is suitable for soft clay, silty clay, loose sandy soil, embankments, foundations of low-to medium rise buildings.

5. Explain in detail the factors that should be considered for ideal bridge site location and ideal bridge type for a particular location. [10]

Answer:

The selection of a bridge site and the type of bridge is one of the most important decisions in bridge engineering. A suitable site minimizes construction and maintenance costs while ensuring safety, durability, and efficient transportation.

There are several factors that should be considered for ideal bridge site location. They are as follows:

i. River characteristics:

- Bridge should cross the river at a straight reach rather than at bends.
- Avoid meandering sections to reduce the risk of bank erosion and scour.
- River width must be as small as possible to reduce bridge length.

ii. Topography:

- Site must be stable and high river banks.
- Deep valleys or steep approaches should be avoided.
- Narrow channel widths and stable banks are chosen to reduce required span and amount of river training works.

iii. Geological conditions:

- The sites with shallow, firm, non-erodible strata are preferred for economical and reliable foundations.
- The bed material, scour potential, and presence of boulders, voids, or weak alluvium with boreholes are investigated and geotechnical tests are carried out.

iv. **Hydraulic conditions:**

- A straight, uniform river reach with steady flow must be chosen to reduce hydraulic complications and scour risk.
- Sufficient vertical and lateral clearance above the highest flood level (HFL) should be ensured to avoid overtopping and debris impact.
- The sites near confluences or abrupt channel changes must be avoided and check historical flood records and sedimentation patterns.

v. **Environmental and social factors:**

- Locations that cause major disruption to ecosystems (wetlands, protected habitats), cultural heritage sites must be avoided.
- The effects on local communities such as flooding patterns, or livelihoods (fisheries, irrigation) and include mitigation in site choice.

vi. **Economic considerations:**

- Overall life-cycle cost (construction plus maintenance, river training, and scour protection) must be evaluated.
- The availability and cost of construction materials (stone, sand, steel, cement) and skilled

Factors for selecting an ideal bridge type:

i. **Span length of the bridge:**

For short span bridges: RCC slab bridge, beam bridge

For medium span bridges: RCC girder bridge, steel girder bridge, prestressed concrete bridges.

For long span bridges: Arch bridge, cable-stayed bridge.

For very long span bridges: suspension bridge

ii. **Foundation:**

Arch bridges and heavy bridges require strong foundation.

Lightweight bridges such as steel truss or cable-supported bridges can be provided with weak foundation.

iii. **Availability of materials:**

RCC bridges are constructed where cement and aggregates are readily available.

Steel bridges are constructed where steel fabrication facilities exist.

Timber bridges can be provided for temporary bridge or rural crossings.

iv. **Economy:**

The bridge must be constructed at low initial cost and have low maintenance cost.

Bridge must have long service life.

There must be ease of inspection and repair.

v. **Seismic and climatic conditions:**

In earthquake-prone regions such as Nepal, bridges should have ductile detailing and seismic-resistant design.

Wind effects are very important for long-span bridges.

Temperature variations and corrosion should also be considered.

Section-C

6. **How is a storm hydrograph different from an annual hydrograph? Sketch a typical storm hydrograph and explain its significance, describing the various parts. [2+3]**

Answer:

A storm hydrograph is a short-term response to one storm, usually measured over hours or day.

An annual hydrograph is the while long-term pattern over 12 months, showing seasonal variation in discharge.

A storm hydrograph shows how a river's discharge changes during and after a single rainfall event, while an annual hydrograph shows how discharge changes over an entire year. The storm hydrograph is used to study flood response.

Storm hydrograph	Annual hydrograph
a. Represents the runoff due to a single storm event.	a. Represents variation in river discharge over one year.
b. Duration is hours to a few days.	b. Duration is 365 days (or one hydrological year).
c. Used for flood estimation, drainage, and spillway design.	c. Used for water resources planning, reservoir operation, and irrigation scheduling.
d. Shows rapid rise and recession of flow.	d. Shows seasonal variations in flow due to rainfall, snowmelt, and dry periods.

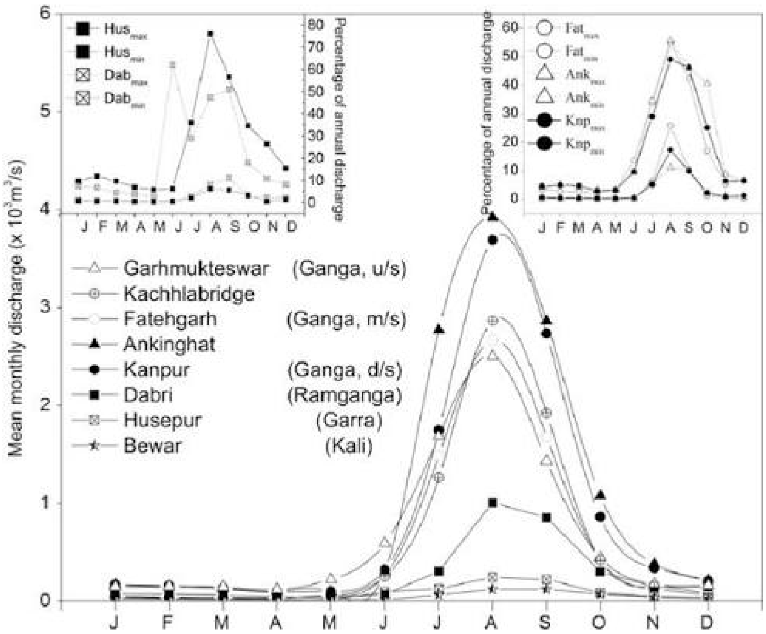
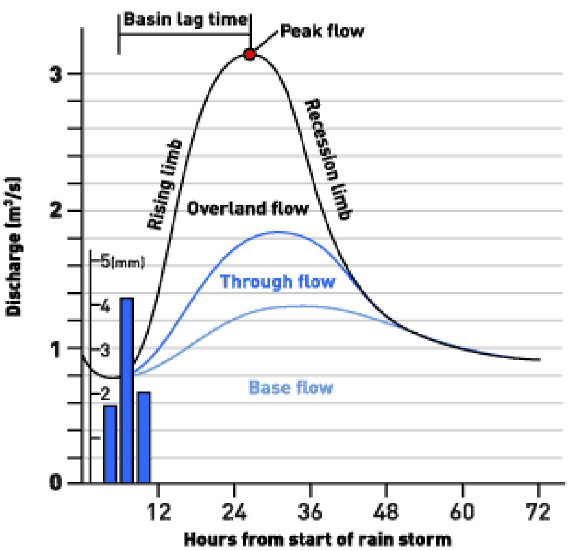


Fig: Annual hydrograph

Typical storm hydrograph:



A typical storm hydrograph has these parts:

a. **Rising limb.**
The steep upward part of the curve after rainfall begins.
The discharge increases as runoff reaches the river.
Indicates the rapid increase in discharge due to rainfall.

b. **Peak discharge.**
The highest point on the hydrograph.
this is the maximum river flow during the storm.

c. **Lag time.**
The time gap between peak rainfall and peak discharge.
A short lag time usually means a flashier flood response.

d. **Recession or falling limb.**
The descending part after the peak, showing discharge gradually returning to normal as runoff decreases.
Represents the gradual decrease in streamflow after rainfall stops.

e. **Base flow.**
The normal groundwater-fed flow in the channel before and after the storm.
Exists even in the absence of rainfall.

f. **Storm flow.**
The extra flow added to the river because of the rainfall event.

g. **Direct runoff**:
It is the portion of runoff generated directly by the storm.
It is the area above the base flow line.

Significance:

A storm hydrograph is very important in hydrology because it is used to:

a. Estimate peak flood discharge.
b. Design spillways, culverts, bridges, and storm drains.
c. Determine runoff volume.
d. Estimate flood arrival time.
e. Study watershed response to rainfall.
f. Develop unit hydrographs.
g. Plan flood forecasting and warning systems.
h. Design reservoirs and detention basins.
i. Analyze the effects of land-use changes on runoff.
j. Assess watershed characteristics such as infiltration and storage.

7. How do you differentiate laminar, transitional and turbulent flow in open channel and pipe flow? Mention limitations of Bernoulli's equation. [4+1]

Answer:

The nature of fluid flow is classified as laminar, transitional, or turbulent based on the value of the Reynolds number (Re).

In pipe flow:

Flow Type	Reynolds Number (Re)
Laminar	($Re < 2000$)
Transitional	($2000 \leq Re \leq 4000$)
Turbulent	($Re > 4000$)

Laminar Flow	Transitional Flow	Turbulent Flow
In laminar flow, fluid flows in smooth, parallel layers without mixing.	In transition flow, fluid flows in intermediate stage between laminar and turbulent flow.	In turbulent flow, fluid particles move randomly with eddies and vortices.
Re < 2000 in laminar flow	**2000 ≤ Re ≤ 4000 in transitional flow.**	**Re > 4000 in turbulent flow.**
The flow in laminar flow is smooth and orderly	The flow in transition flow is unstable and fluctuating	The flow in this turbulent flow is highly irregular and chaotic
Mixing of fluid in laminar flow is negligible.	Mixing of the fluid in transition flow is partial mixing.	Mixing of the fluid in turbulent flow is intense mixing
Dominant force in laminar flow is viscous force.	Dominant force in transitional flow is both viscous and inertial forces	Dominant forces in turbulent flow is inertial forces.

In open channel flow:

Flow Type	Reynolds Number (Re)
Laminar	(Re < 500)
Transitional	$(500 \le Re \le 2000)$
Turbulent	(Re > 2000)

Laminar Flow	Transitional Flow	Turbulent Flow
Reynolds number for the laminar flow is **< 500**	Reynolds number for the transitional flow is **500 ≤ Re ≤ 2000**	Reynolds number for the turbulent flow is **Re > 2000**
In laminar flow, the fluid flow occurs in smooth, parallel layers.	In turbulent flow, fluid flow is partly laminar and partly turbulent.	In turbulent flow, fluid flow is irregular with random motion and eddies.
Mixing of water particles is negligible in laminar flow.	Moderate mixing occurs in transitional flow.	Intense mixing of water particles occurs in turbulent flow in open channel..
Flow is stable and orderly.	Flow is unstable and easily disturbed.	Flow is chaotic but statistically stable.
Rare in natural rivers and canals; mainly found in laboratory channels or very slow flow.	Occurs only over a narrow range of flow conditions.	Common in rivers, canals, spillways, and drainage channels.
Occurs at very low velocities.	Occurs at intermediate velocities.	Occurs at high velocities.

Bernoulli's equation is based on the conservation of mechanical energy and is applicable only under certain assumptions. The major limitations of Bernoulli's equation are as follows:

1. **Applicable only to steady flow**:

   The flow parameters (velocity, pressure, and discharge) must not change with time.

2. **Valid only for incompressible fluids**:

   The fluid density must remain constant.
   It is generally applicable to liquids like water but not to high-speed compressible gas flows.

8. Define the types of irrigation. Explain the suitability of those types in the context of Nepal's topography. [2+3]

Answer:

Irrigation is an artificial application of water to agricultural land in the required quantity and at the required time to supplement natural rainfall for optimum crop growth and yield.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of irrigation are briefly described along with its advantages, limitations and suitability.

i. **Surface irrigation:**

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.
- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Suitability:
    - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
    - This method is suitable for row and field crops.

Surface irrigation is further classified into:

a. Flooding method: entire land surface is covered with water and allowed to stand on the field.
b. Contour farming: adapted to hilly areas with steep slopes and quickly falling contours.
c. Furrow irrigation: common method of irrigation where water is applied in small streams between the rows of the crops, grown on the ridges or furrows.

ii. **Sprinkler irrigation:**

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Suitability:
    - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
    - It is suitable for closely spaced crops like cereals and fodders.

iii. **Drip irrigation:**

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- **Suitability:**
    - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
    - The method is ideal for row crops, orchard and vineyards.

9. **a. How do you calculate reservoir capacity using mass inflow curve and installed capacity of a ROR plant using flow duration curve? Discuss with neat sketch and labels. [3+3]**
b. Define optimization. How do you optimize installed capacity of a ROR plant using marginal cost and marginal benefit method? Discuss. [1+3]
Answer:

A mass inflow curve is a graph showing the cumulative inflow volume into a reservoir plotted against time. It is used to determine the storage (reservoir) capacity required to meet a constant water demand.

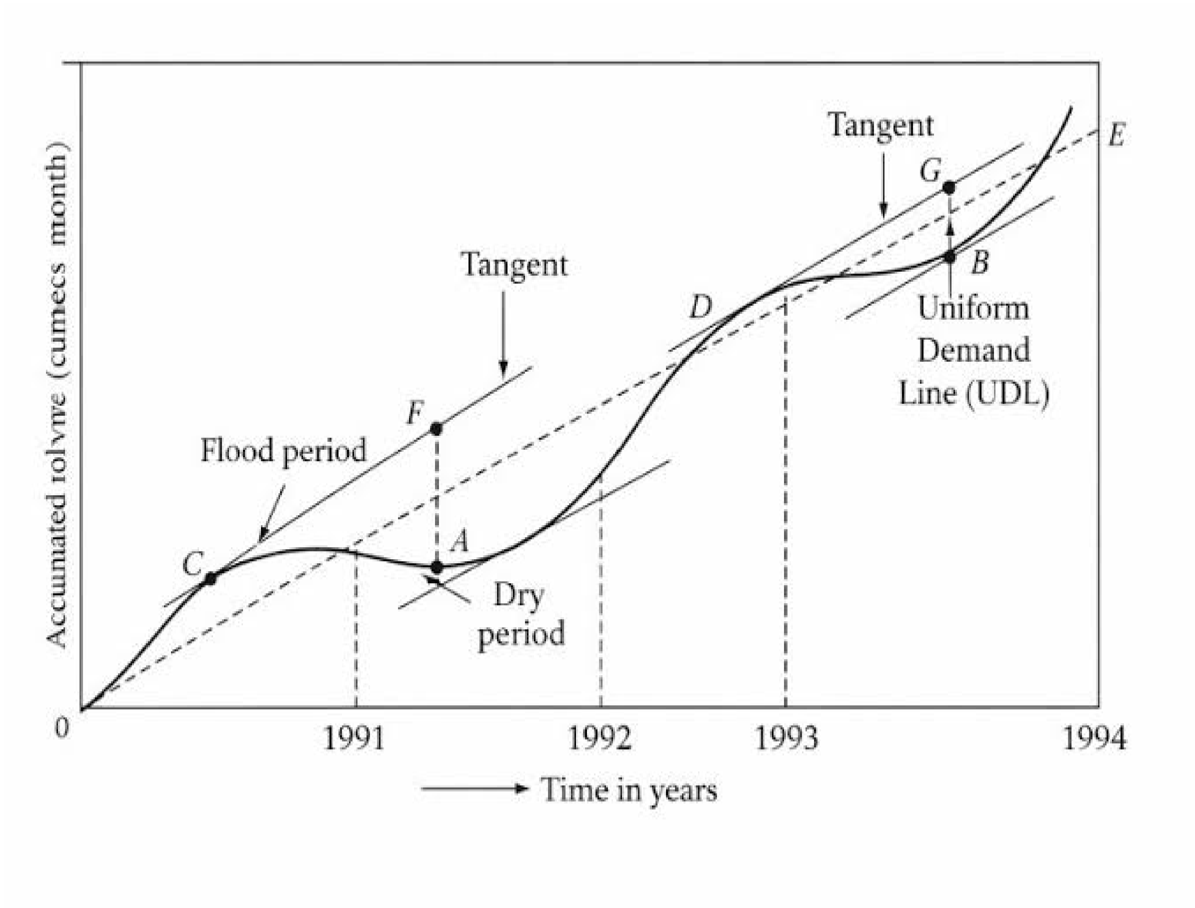
Principle:

- If the cumulative inflow is greater than the cumulative demand, water is stored.
- If the cumulative demand exceeds cumulative inflow, water is supplied from storage.
- The maximum difference between the cumulative demand line and the mass inflow curve gives the required reservoir capacity.

Procedure:

d. The monthly or daily streamflow data is collected.
e. The cumulative inflow volume is computed.
f. The cumulative inflow (Y-axis) versus time (X-axis) is plotted.
g. A straight demand line with the slope equals to constant demand is drawn.
h. Then, the parallel demand lines touching the mass curve is also drawn.
i. The maximum vertical distances between the demand line and the mass curve is measured.

Hence, $\boxed{\text{Reservoir Capacity} = \text{Maximum cumulative deficit}}$



Installed capacity of ROR plant using Flow Duration Curve (FDC)

A Flow Duration Curve (FDC) is a plot showing the percentage of time that a given discharge is equaled or exceeded. It is widely used to determine the design discharge and hence the installed capacity of a Run-of-River hydropower plant.

Hydropower output is given by:

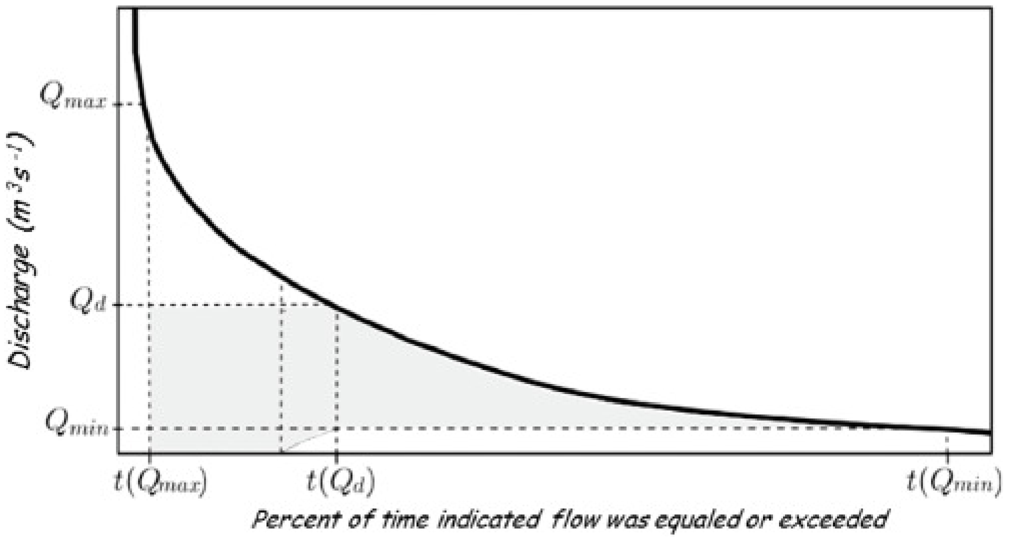
$$\boxed{P = \rho g Q H \eta}$$

$$where:$$
$$P = Power(W)$$
$$\rho = Density\ of\ water(1000kg/m^3)$$
$$g = Acceleration\ due\ to\ gravity(9.81m/s^2)$$
$$Q = Design\ discharge(m^3/s)$$
$$H = Net\ head(m)$$
$$\eta\ =\ Overall\ efficiency$$
$$\boxed{P = \frac{9.81\, QH\eta}{1}}$$

Procedure:

1. Collect long-term streamflow data.
2. Arrange discharges in descending order.
3. Compute exceedance probability.
4. Plot the Flow Duration Curve.
5. Select the design discharge (e.g., Q40, Q50, or Q60).
6. Calculate installed capacity using

$$P = \rho g Q H \eta.$$



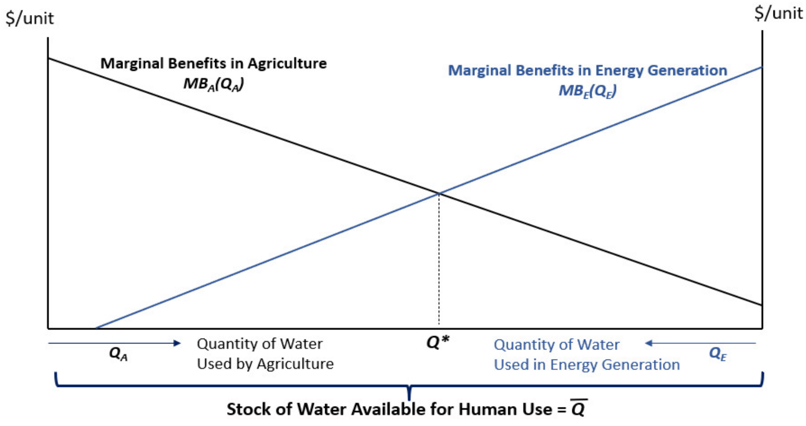
Optimization:

It is the process of determining the best or optimum solution from several feasible alternatives so that a specific objective (such as minimizing cost or maximizing benefit) is achieved while satisfying all technical, economic, environmental, and operational constraints.

In hydropower engineering, optimization of installed capacity means selecting the plant capacity that provides the maximum net economic benefit over the project's life.

Procedure for optimization:

1. Long-term river flow data are collected.
2. By using the data, Flow Duration Curve (FDC) is prepared.
3. Several trials installed capacities are selected.
4. An annual energy generation for each capacity is then estimated.
5. The annual revenue (benefit) is calculated.
6. The capital and operation & maintenance costs is estimated.
7. The marginal cost and marginal benefit for each increment of capacity is then computed.
8. Now the capacity where **Marginal cost = Marginal Benefit** is selected.



Section-D

10. List the requirements of highway drainage system. [5]

Answer:

A highway drainage system is the arrangement of surface and subsurface structures provided to collect, convey, control, and safely dispose water from the roadway and its surrounding area, thereby protecting the pavement, embankment, and road users from water-related damage. Good highway drainage is important as it prevents road surface from being soft and losing its strength and its bearing capacity. It also prevents failure of slope formation and erosion of side slopes.

One of the commendable examples of good highway drainage in Nepal **is** the drainage arrangement along BP highway (Sindhuli road), where combination of cross drainage, side slope drainage and sub surface horizontal drainages are provided to divert the surface and groundwaters from the road surface, which not only manages the torrential monsoon rains but also protect mountainous terrain from water-induced landslides.

Requirements of highway drainage system:

a. The surface water from the carriageway and shoulder should be effectively drained off without allowing it to percolate to subgrade.
b. The surface water from the adjoining land should be prevented from entering the roadway.
c. The side drain should have sufficient capacity and longitudinal slope to carry away all the surface water collected.
d. Flow of the surface water across the road and shoulders and along the slopes should not cause formation of cross ruts and erosion.
e. Seepage and the other sources of underground water should be drained off by subsurface drainage system.
f. Highest level of ground water table should be kept well below the level of subgrade, preferably at least 1.2 m.
g. In water-logged areas special precautions should be taken, especially if detrimental salts are present or if flooding is likely to occur.

11. Explain in brief various design factors to be considered in pavement design. [10]

Answer:
The main objective of pavement design is to provide a durable, safe, and economical pavement capable of carrying the anticipated traffic loads throughout its design life.
There are various design factors that are explained in brief:

a. Traffic loading:

- Higher traffic loads require thicker and stronger pavement layers.
- Traffic loading depends on:
  - Magnitude and frequency of wheel loads.
  - Number of commercial vehicles
  - Axle load configuration (single, tandem, tridem).
  - Traffic growth rate.
  - Cumulative standard axles (CSA) or Equivalent Single Axle Loads (ESAL).

b. Subgrade soil characteristics:

- Weak subgrade requires improved soil or thicker pavement.

- It depends on:
    - Soil type and classification.
    - Bearing capacity (CBR for flexible pavement).
    - Modulus of subgrade reaction (k-value) for rigid pavement.
    - Compressibility and settlement characteristics.
    - Swelling and shrinkage potential.

c. Climate and environmental conditions:

- Climate influences pavement strength, cracking, rutting, and durability.
- It depends on:
    - Rainfall and drainage conditions.
    - Temperature variation.
    - Freeze–thaw cycles (cold regions).
    - Moisture fluctuation.
    - Solar radiation and humidity.

d. Drainage conditions:

- Poor drainage weakens the pavement and shortens its service life.
- It depends on:
    - Surface drainage efficiency.
    - Subsurface drainage.
    - Groundwater level.
    - Water infiltration through pavement joints and cracks.

e. Pavement materials:

- High-quality materials improve pavement performance and longevity.
- It depends on:
    - Quality of aggregates.
    - Bitumen or cement properties.
    - Strength, stiffness, and durability.
    - Availability of local materials.
    - Resistance to weathering and wear.

f. Design life:

- Longer design life generally requires a stronger pavement structure.
- Design life is the expected service period before major rehabilitation.

Common values:
- Flexible pavement: **15–20 years**
- Rigid pavement: **20–30 years**

g. Safety and reliability:

- Higher reliability is adopted for important highways and expressways.
- It is a probability that the pavement performs satisfactorily during its design life.
- Considers uncertainties in traffic prediction, material properties, and construction quality.

h. Construction quality:

- Even a well-designed pavement may fail prematurely if construction quality is poor.
- It depends on:
    - Proper compaction.
    - Adequate layer thickness.
    - Quality control of materials.
    - Good workmanship.

i. **Performance criteria:**
- The pavement must satisfy the following criteria:
  - Strength against repeated loading.
  - Acceptable riding quality.
  - Resistance to rutting.
  - Fatigue resistance.
  - Skid resistance.
  - Serviceability throughout the design life.

j. **Economic considerations:**
- The most economical design is selected while meeting performance requirements.
- It depends on:
  - Initial construction cost.
  - Maintenance and rehabilitation cost.
  - Life-cycle cost analysis.
  - Availability of funds.

k. **Availability of materials and equipment:**
- Readily available resources reduce construction cost and time.
- It includes:
  - Local availability of suitable aggregates, bitumen, cement, and stabilizing materials.
  - Availability of machinery and skilled labor.

12. Name different types of loads coming on the foundation of a super structure. How are foundations classified as? What are causes of foundation failures? [2+4+4]

Answer:

Foundation is the lowermost part of structure which receives load from superstructure and transfers it to stable soil layer. The principal loads are:

1. **Dead Load**:
   - Self-weight of structural components such as slabs, beams, columns, walls, roofs, and the foundation itself.
   - It is a permanent and static load.

2. **Live (Imposed) Load:**
   - Loads due to occupants, furniture, equipment, vehicles, stored materials, etc.
   - These loads vary in magnitude and position.

3. **Wind Load:**
   - Horizontal and uplift forces caused by wind pressure.
   - Significant for tall buildings, towers, and industrial structures.

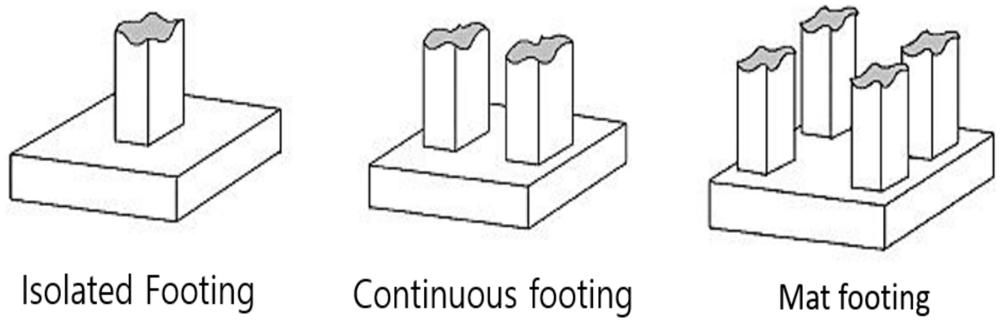
4. **Earthquake (Seismic) Load**:
   - Dynamic forces induced during earthquakes.
   - Causes horizontal and vertical ground motions that affect the foundation.

5. **Impact and Vibratory Loads**:
   - Produced by machinery, cranes, moving vehicles, and dynamic industrial equipment.

Foundations are broadly classified into two categories:

a. **Shallow foundations:**

➢ Used when the soil with adequate bearing capacity is available near the ground surface (generally depth ≤ width).



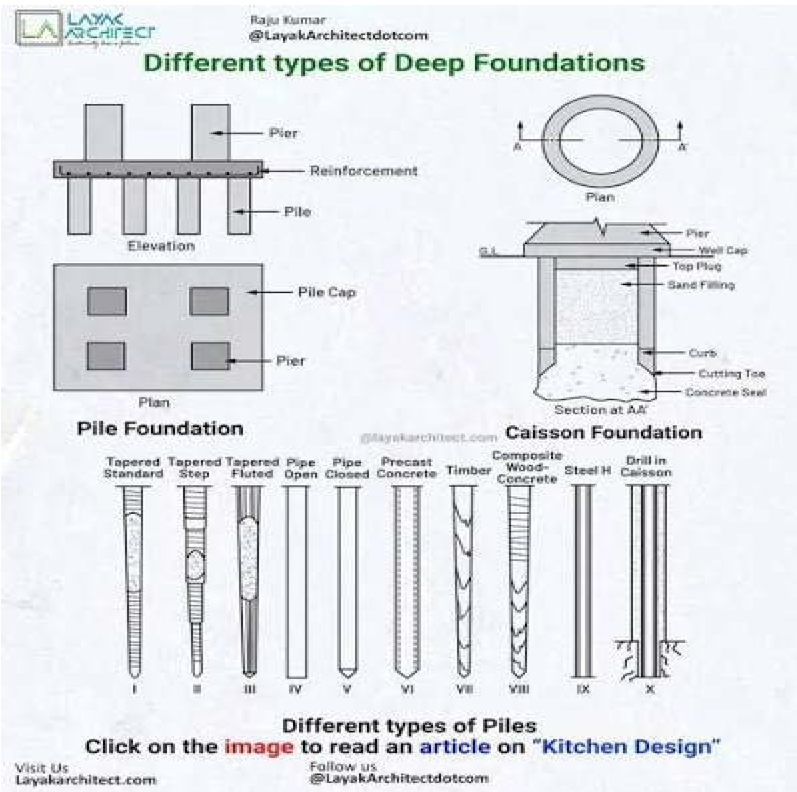
Types:

a. Strip (Wall) Footing
b. Isolated (Pad) Footing
c. Combined Footing
d. Strap (Cantilever) Footing
e. Continuous Footing
f. Raft (Mat) Foundation

Applications: Residential buildings, low-rise structures, and light to moderate loads.

b. Deep foundations:

Used when the upper soil is weak and stronger strata are available at greater depths (depth > width).



Types:

- Pile Foundation

- Pier Foundation
- Caisson (Well) Foundation

Applications: High-rise buildings, bridges, waterfront structures, and heavy industrial facilities.

Causes of foundation failures:

a. **Insufficient Bearing Capacity**:

Soil fails due to excessive structural load.

b. **Excessive Settlement**:

Total settlement exceeds permissible limits.

c. **Differential Settlement**:

Unequal settlement causes cracks, tilting, and structural distortion.

d. **Poor Soil Investigation**:

Inadequate geotechnical exploration leads to improper foundation design.

e. **Water-Related Problems**:

Poor drainage, seepage, erosion, scour, or fluctuating groundwater weakens the soil.

f. **Poor Design**:

Incorrect estimation of loads, inadequate footing size, or insufficient reinforcement.

g. **Poor Construction Quality**:

Improper excavation, inadequate compaction, poor concrete quality, or faulty workmanship.

h. **Soil Volume Changes**:

Swelling and shrinkage of expansive soils due to moisture variations.

i. **Earthquake and Dynamic Effects**:

Soil liquefaction, excessive vibration, or seismic loading can damage foundations.

j. **Adjacent Excavation or Construction**:

Nearby excavations may remove lateral support, causing settlement or instability.

k. **Overloading**:

Addition of extra storeys or increased loads beyond the original design capacity.

l. **Chemical Attack and Corrosion**:

Sulphates, chlorides, or aggressive groundwater may deteriorate concrete and reinforcement.

Section-E

13. The following observations were made on a sewage sample at 20 °C. [5]

$BOD_{4,\,20°C}$ = 290.50 mg/l
$BOD_{1,20°C}$=56.15 mg/l
Calculate rate reaction constant k and ultimate first stage BOD at 25°C.
Answer:

$$For\ first - stage\ BOD, the\ base - 10\ equation\ is:$$

$$\mathrm{BOD}_t = L_0(1 - 10^{-kt})$$

$$where:$$

$$L_0 = Ultimate\ first - stage\ BOD(mg/L)$$

$$k = deoxygenation\ \ rate\ constant\ (day^{-1})$$

$$t = time\ (days)$$

$$\mathrm{BOD}_1 = 56.15\ \mathrm{mg}/L\ at\ 20^{\circ\mathrm{C}}$$

$$\mathrm{BOD}_4 = 290.50\ \mathrm{mg}/L\ at\ 20^{\circ\mathrm{C}}$$

Using,

$$\frac{1 - 10^{-4k}}{1 - 10^{-k}} = \frac{290.50}{56.15} = 5.17361$$

Let, $x = 10^{-k}$

Then solving,

$$x \approx 0.774x \approx 0.774x \approx 0.774$$

Hence,

$$k_{20} = -\log_{10}(0.774) \approx 0.111\ \mathrm{day}^{-1}$$

Again,

$$L0 = \frac{\mathrm{BOD}_1}{1 - 10^{-k}} = \frac{56.15}{1 - 0.774} = \frac{56.15}{0.226} \approx 248.5\ mg/L$$

Temperature correction:

$$\begin{gathered} k_T = k_{20}(1.047)^{T-20} = 0.111(1.047)^5 \\ \approx (1.047)^5 \approx 1.258 \\ = 0.111 \times 1.258 \\ k_{25} \approx 0.140\ \mathrm{day}^{-1} \\ \boxed{k_{25} \approx 0.140\ \mathrm{day}^{-1}} \end{gathered}$$

$$The\ \boldsymbol{ultimate\ first - stage\ BOD}\ is\ independent\ of\ temperature, so$$
$$\boxed{L_0 = 248.5\ \mathrm{mg/L}}$$

14. Briefly differentiate between Initial Environment Examination (IEE) and Environmental Impact Assessment (EIA) with their context and applications. [5]

Answer:

IEE is the study conducted to determine whether a proposed medium scale project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts. Similarly, EIA is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.

The difference between IEE and EIA are as follows:

IEE	EIA
i. The scope of IEE is limited and less detailed.	i. The scope of EIA is extensive, detailed and comprehensive.
ii. IF IEE is approved, EIA is not required.	ii. If IEE is not approved, EIA is required.
iii. The notice can be published in a **local newspaper.**	iii. The notice must be published in a **national daily newspaper.**
iv. IEE covers small- and medium-scale projects with limited environmental impacts.	iv. EIA covers large-scale or environmentally sensitive projects with potentially significant impacts.
v. The approval must be granted within **15 days.**	v. The concerned body has **35 days** to approve the report after receiving suggestions.
vi. Public participation is limited or not always mandatory.	vi. Public participation is mandatory.
vii. IEE is done for: a. Highways: 5-50 km b. Irrigation projects with CA= 25-2000 ha c. Hydropower projects with installed capacity= 1-50 MW	vii. EIA is done for: a. Highways: >50 km b. Irrigation projects with CA= >2000 ha c. Hydropower projects with installed capacity= >50 MW

15. What are the major factors governing in the water demand and variation of water demand? Briefly describe in context of Nepal. [10]

Answer:

Water demand is necessary for designing the water supply schemes. There are different types of water demand such as domectic demand, livestock demand, municipal demand etc. In Nepal, domestic demand in fully plumbed house is taken as 112 lpcd and for partially plumbed house is taken as 65 lpcd. Similarly, livestock demand is taken as 20% of the domestic demand and so on. As so many factors are involved in the demand of water, it is not possible to accurately determine the actual demand. Some of the factors governing the water demand are described below:

i. Population:

Water demand increases with population size and population growth.
Future population projection is essential for designing water supply systems.

ii. Climate:

Hot and dry climates increase water consumption for drinking, bathing, and irrigation.
Cooler climates generally require less water.

iii. Standard of living:

Higher living standards lead to greater use of water through showers, washing machines, gardens, and household appliances.

iv. Quality and availability of water:

A reliable supply of safe drinking water encourages higher consumption.
Scarcity or poor quality of water reduces demand.

v. Metering of water supply:

Metered connections reduce wastage and promote efficient water use.
Unmetered supply often leads to excessive consumption.

In Nepal, water demand is influenced by unique geographical, climatic, and socio-economic conditions:

- Rapid urbanization in cities such as Kathmandu, Pokhara, and Biratnagar has significantly increased domestic water demand.
- Seasonal variation is increased with water scarcity common during the dry season (March–May) because many springs and streams experience reduced flow.
- Increased population growth rate in urban areas have placed additional pressure on existing water supply systems.
- In many municipalities, intermittent water supply and leakage in distribution networks lead to water losses and uneven availability.
- Industrial development and the growth of the tourism sector have further increased water requirements.
- In rural and hilly regions, dependence on springs and gravity-fed systems means that climate change and drying water sources directly affect water availability and demand.

Hence, improvement of rainwater harvesting, source protection, leakage control, and efficient water management is increasingly important to meet future water demand.

Variation of water demand:

i. **Seasonal variation:**

Demand increases during summer due to higher temperatures.
It decreases during the winter season.
In Nepal, tourist seasons may also increase demand in cities like Pokhara and Kathmandu.

ii. **Daily variation:**

Water consumption is generally higher on weekends, festivals, and holidays than on ordinary working days.

iii. **Hourly variation:**

Peak demand occurs in the morning (6–9 AM) and evening (6–9 PM) due to bathing, cooking, and household activities.
Demand is lowest during late night hours.

iv. **Emergency variation:**

Sudden increases occur due to fire-fighting, drought, pipe bursts, festivals, or natural disasters.